LAB 6

Introduction to Robot Operating System (ROS)

Student Details

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1. Aim

To install and verify ROS 2 Humble, MoveIt, RViz, PlanSys2, and Gazebo, then run the Panda arm pick-and-place demonstration using MoveIt Task Constructor and PlanSys2.

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2. Software/Environment Used

- Operating System: Ubuntu 22.04 / WSL2 / Mac (as applicable)
- ROS 2 Humble
- MoveIt 2
- RViz2
- PlanSys2
- Gazebo / Ignition Fortress
- Panda robot simulation/demo packages

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3. Part A: Environment and Software Installations

3.1 ROS 2 Humble Installation Verification

Command used:

```
source /opt/ros/humble/setup.bash
ros2 --help
ros2 pkg list
```

Result/Output:

```
administrator@cs-pro:~/Desktop/Folders/U23CS014$ ros2 --help
usage: ros2 [-h] [--use-python-default-buffering] Call `ros2 <command> -h` for more detailed u

ros2 is an extensible command-line tool for ROS 2.

options:
  -h, --help            show this help message and exit
  --use-python-default-buffering
                        Do not force line buffering in stdout and instead use the python defau
                        stdout is interactive or not
```

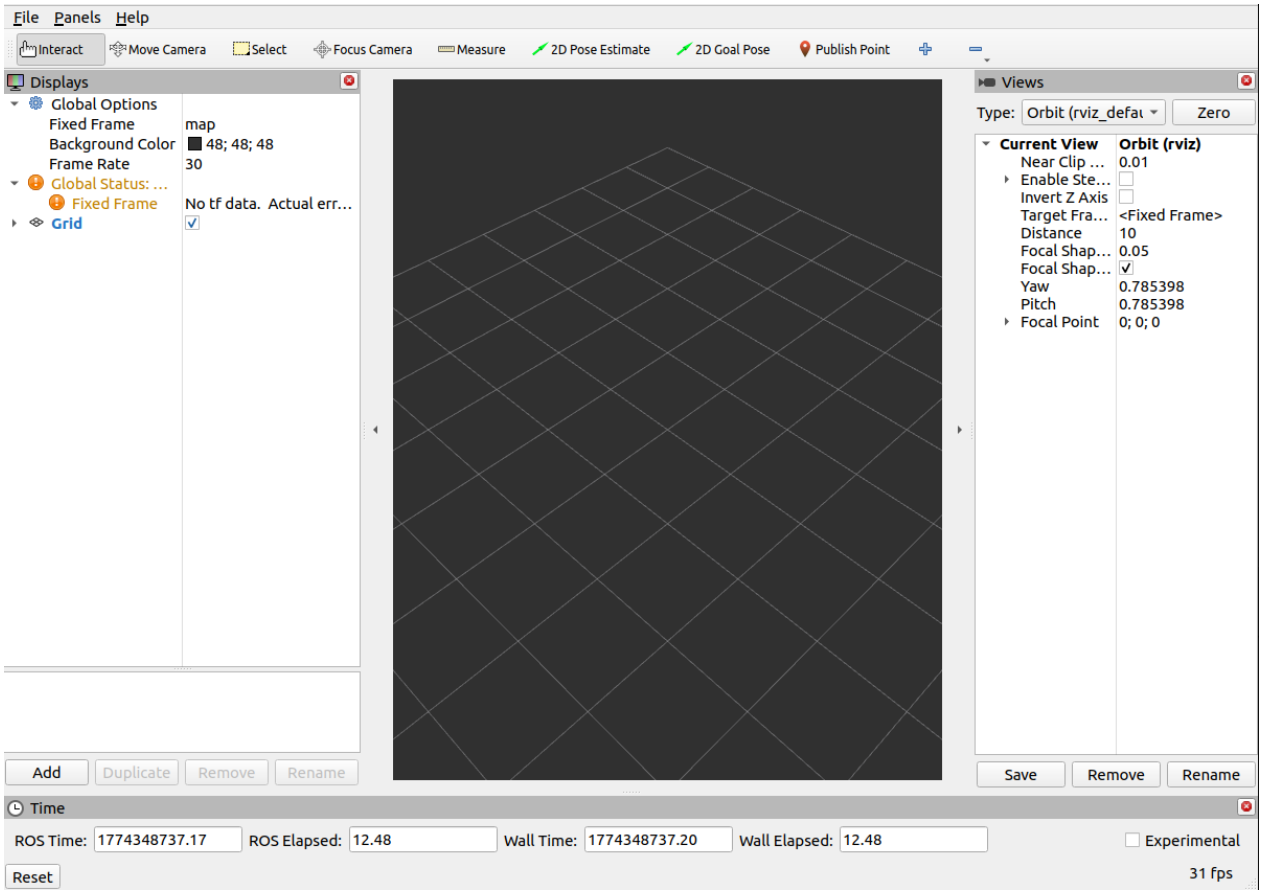
Observation/Comment: ROS 2 Humble was installed and configured correctly, and the terminal recognized ROS 2 commands after sourcing the setup file.

3.2 RViz2 Verification

Command used:

```
source /opt/ros/humble/setup.bash
ros2 run rviz2 rviz2
```

Result/Output:



alt text

Observation/Comment: RViz2 started properly and can be used for robot visualization, planning, and scene inspection.

3.3 PlanSys2 Installation Verification

Command used:

```
sudo apt install ros-humble-plansys2-*
```

Result/Output:

- PlanSys2 packages were installed successfully.
- Required dependencies were resolved.

[Output too long to be shown]

3.4 Gazebo / Ignition Fortress Installation Verification

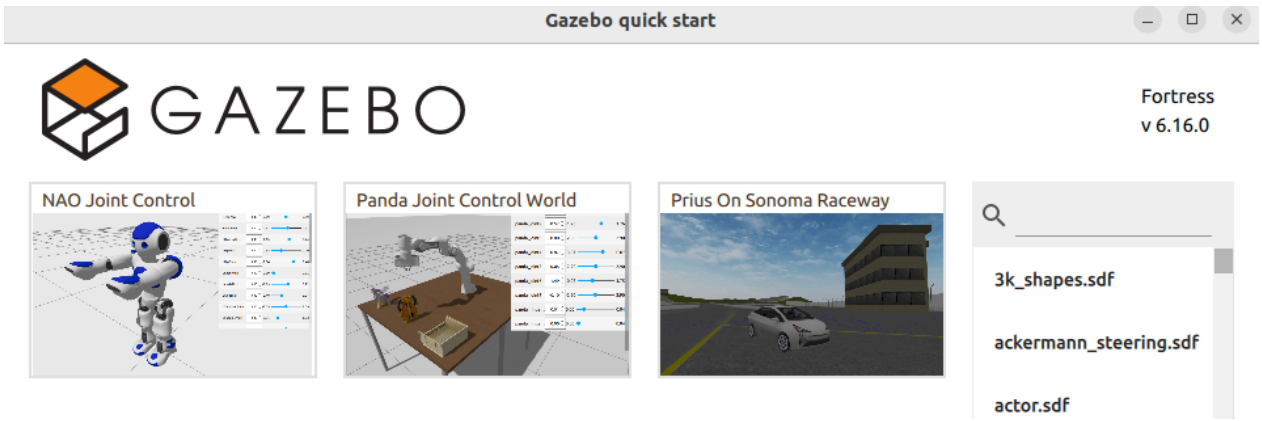
Commands used:

```
sudo apt-get update
sudo apt-get install lsb-release gnupg
sudo curl https://packages.osrfoundation.org/gazebo.gpg --output /usr/share/keyrings/pkgs-osr

echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/pkgs-osrf-archive-
sudo apt-get update
sudo apt-get install ignition-fortress
```

Result/Output:

- Gazebo repository was added.
- Ignition Fortress installation completed successfully.



Gazebo-Fortress

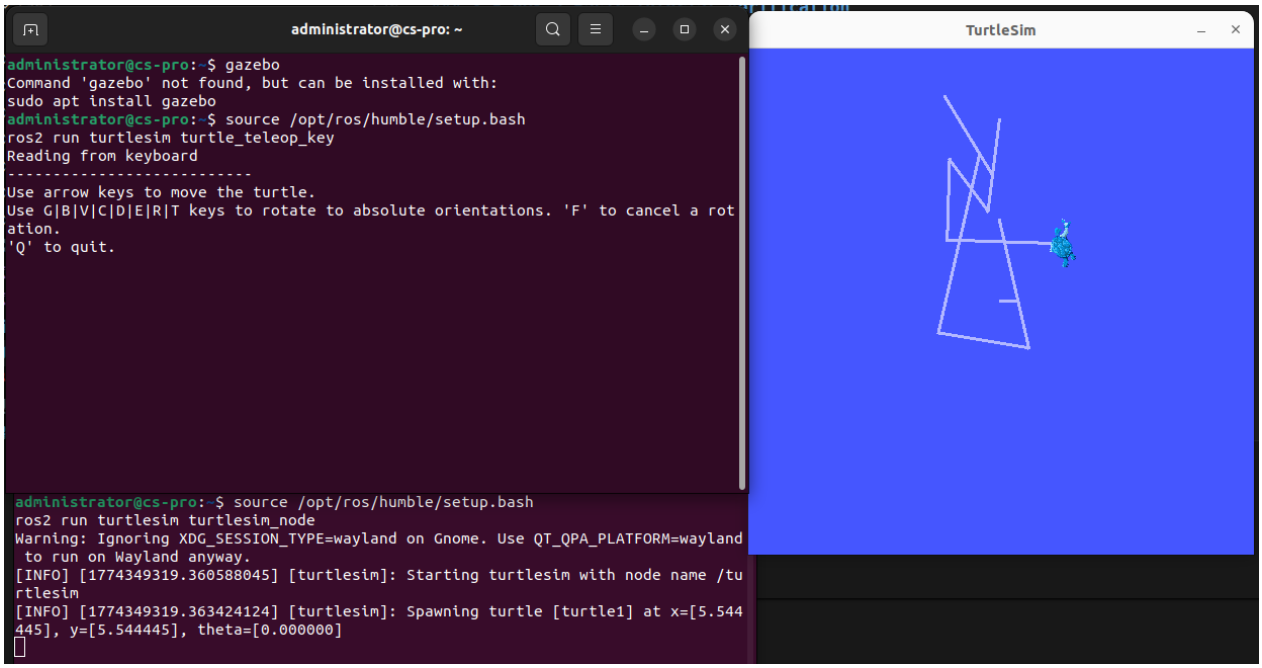
Observation/Comment: Gazebo/Ignition Fortress was installed and can be used for simulation and robot testing.

3.5 ROS 2 Basic Tutorial Verification

Commands used:

```
cd ~/ros2_ws/src
ros2 pkg create --build-type ament_cmake <package_name>
ros2 pkg list
ros2 pkg executables turtlesim
ros2 run turtlesim turtlesim_node #T1
ros2 run turtlesim turtle_teleop_key #T2
```

Result/Output:



Ttl Sim and teleop

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4. Part A: MoveIt Installation and Verification

4.1 MoveIt Task Constructor Repository Setup

Commands used:

```
cd ~/ws_moveit/src
git clone -b humble https://github.com/moveit/moveit_task_constructor.git
cd ~/ws_moveit
```

5. Part B: Panda Arm Demo

```
# Terminal 1:
source install/setup.bash
ros2 launch moveit2_tutorials mtc_demo.launch.py # Loads the robot

# Terminal 2
source install/setup.bash
ros2 launch moveit2_tutorials pick_place_demo.launch.py # Pick and Place operation
```

Result/Output:

Robot
Robot

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5. Part B: Panda Arm Pick-and-Place

Source Code Setup

Source used:

- PANDA_PICK_PLACE / AVCAR_PANDA_PICK_PLACE project files

Commands used:

```
colcon build
source install/setup.bash
```

5.1 Launching the Panda Pick-and-Place Demo

Commands used: Terminal 1:

```
ros2 launch plansys2_bt_example plansys2_bt_example_launch.py
```

Terminal 2:

```
ros2 launch panda_ros2_moveit2 panda_interface.launch.py
```

Terminal 3:

```
ros2 run ros2_clients actor
```

Terminal 4:

```
ros2 run plansys2_bt_example server
```

Result/Output:

- The demo components launched successfully.
- The Panda interface, PlanSys2 server, and supporting node(s) started.
- Plan and execution behavior was visible in the interface/logs.

Observation/Comment: - The pick-and-place system launched correctly. The arm was able to follow the planning sequence when a valid plan was generated. - The pick-and-place workflow followed the expected sequence of pre-grasp, approach, grasp, lift, transfer, place, and retreat.